KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Restaurant Order Processing and Billing System

Submitted by

JEEVA R

25BIT022

Under the Guidance of

DHILEEBAN A

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026 Semester:I

Semester: **I**

DECLARATION

I hereby declare that the project entitled "**Restaurant Order Processing and Billing System"** submitted in partial fulfillment of the requirements for the course [JAVA PROGRAMMING**]** is a **record of my original work** carried out under the guidance of **[DHILEEBAN A]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

JEEVA R

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Restaurant Order Processing and Billing System"** is a **bona fide work** carried out by **[JEEVA R] [I-BSc IT-'A']**, , **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09/04/2026

Submitted for the Viva-Voce Examination held on **09/04/2026**

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The **Restaurant Order Processing and Billing System** is a Java-based application designed to automate the billing process in restaurants and hotels. In many small restaurants, billing operations are performed manually, which can lead to calculation errors, delays in service, and inefficient record management. This project aims to simplify the billing process by providing an automated system that calculates the total cost of ordered food items and generates an accurate invoice.

The system allows users to select food items from a predefined menu, enter the quantity of each item, and automatically calculate the total bill including applicable taxes such as GST. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface**.

The main objective of the system is to improve billing efficiency, reduce human errors, and provide a faster service to customers. The system generates a clear and structured bill that displays the ordered items, quantities, prices, subtotal, tax, and final payable amount.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced restaurant management systems in the future.

1. INTRODUCTION

In the hospitality industry, efficient billing systems play a crucial role in managing restaurant operations and ensuring customer satisfaction. Traditionally, many small restaurants and hotels rely on manual billing methods, where staff calculate the total bill using handwritten notes or simple calculators. This manual process is time-consuming and can often lead to calculation errors, delays in service, and difficulty in maintaining proper billing records.

With the advancement of computer technology, automated billing systems have become essential for improving efficiency and accuracy in restaurant management. A computerized billing system helps restaurant staff quickly generate bills, calculate totals automatically, and maintain organized records of orders and transactions.

The **Hotel & Restaurant Billing System** is a Java-based application designed to automate the process of order management and bill generation. The system allows the user to select food items from a predefined menu, enter the quantity of items ordered, and automatically calculate the total bill including applicable taxes. The application generates a clear and structured invoice displaying the ordered items, quantities, prices, subtotal, tax, and final payable amount.

This project is developed using **Core Java programming concepts** such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a **console-based program or using Java Swing for a graphical user interface (GUI)**.

The primary goal of this system is to reduce manual effort, improve billing accuracy, and provide faster service to customers. This project also demonstrates how Java programming can be applied to develop simple real-world business applications.

1.1 Objective

The main objective of the **Hotel & Restaurant Billing System** is to develop a simple and efficient billing application using Core Java that automates the process of order management and bill generation in restaurants and hotels.

The specific objectives of this project are:

- To design and develop a **Java-based restaurant billing system**.
- To automate the **calculation of food item prices and total bill**.
- To reduce **manual errors in billing calculations**.
- To provide a **user-friendly system for restaurant staff** to generate invoices quickly.
- To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements.

- To generate a **clear and organized bill for customers**.

1.2 Overview

The **Hotel & Restaurant Billing System** is a menu-driven application developed using Java programming language. The system provides a digital platform where restaurant staff can select food items from a predefined menu and generate bills automatically.

The system displays a list of food items along with their prices. The user can select the required items and enter the quantity ordered by the customer. The program calculates the total price for each item and generates a final bill including applicable taxes such as GST.

The system helps simplify the billing process by eliminating manual calculations and reducing the chances of billing errors. The application can be developed as a **console-based program or a graphical user interface using Java Swing**.

This project demonstrates the practical use of Java programming in developing real-world applications for small business environments such as restaurants and hotels.

1.3 Features

The **Hotel & Restaurant Billing System** includes the following main features:

• **Menu Display** – Displays a list of food items available in the restaurant along with their prices.

• **Item Selection** – Allows the user to select food items from the menu and enter the quantity ordered by the customer.

• **Automatic Bill Calculation** – Automatically calculates the total amount based on the selected items and quantities.

• **Tax Calculation** – Calculates applicable taxes such as GST and adds them to the total bill.

• **Invoice Generation** – Generates a detailed bill showing item names, quantities, prices, subtotal, tax, and final amount.

• **User-Friendly Interface** – Provides a simple and easy-to-use menu-driven interface.

• **Fast Billing Process** – Reduces the time required to generate customer bills.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many small restaurants and hotels, billing is often performed manually using paper or simple calculators. This traditional method creates several challenges.

Some common problems include:

- Errors in bill calculation
- Time-consuming billing process
- Difficulty in managing multiple orders
- Lack of proper billing records
- Inefficient service during busy hours

Manual billing systems also increase the workload for restaurant staff and can lead to customer dissatisfaction due to delays or incorrect billing amounts.

Therefore, there is a need for an automated billing system that can quickly calculate the total bill, manage food orders, and generate accurate invoices.

The **Hotel & Restaurant Billing System** is designed to solve these problems by providing a computerized solution that automates the billing process.

2.2 Project Overview

The **Hotel & Restaurant Billing System** is a Java-based application developed to simplify the billing process in restaurants and hotels.

The system provides a menu-driven interface where users can select food items from a predefined list and specify the quantity of each item ordered by the customer. The system automatically calculates the total price of the selected items and generates a final bill.

The system includes the following functionalities:

• Displaying the restaurant menu with item prices
• Selecting food items and entering quantity
• Calculating item cost automatically
• Computing subtotal and tax amount
• Generating the final bill for the customer

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The scope of the **Hotel & Restaurant Billing System** includes the following:

• Managing restaurant menu items
• Processing customer food orders
• Calculating the total bill automatically
• Generating a detailed invoice for the customer
• Demonstrating the practical use of Java programming concepts

The system can be used in:

• Restaurants
• Hotels
• Cafeterias
• Food courts
• Small food businesses

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Hotel & Restaurant Billing System** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Hotel & Restaurant Billing System** are explained below:

1. Menu Display Module

This module is responsible for displaying the list of available food items along with their prices. The menu helps the user select the required items for billing.

Example Menu:

1. Veg Fried Rice ₹120
2. Chicken Biryani ₹200
3. Paneer Butter Masala ₹180
4. Butter Naan ₹40
5. Cold Drink ₹30

The menu is displayed when the program starts so that the user can easily view the available options.

2. Order Selection Module

This module allows the user to select food items from the menu and enter the quantity ordered by the customer.

Functions of this module include:

• Accepting user input for item selection
• Accepting quantity of selected items
• Storing order details temporarily

The system allows users to select multiple food items before generating the final bill.

3. Bill Calculation Module

This module performs all the billing calculations.

The system calculates:

• Total price of each item
• Subtotal of all selected items
• Applicable tax (GST)
• Final payable amount

The calculation is performed automatically using Java programming logic.

Example Calculation:

Item Total = Price × Quantity
GST = 5% of Subtotal
Final Bill = Subtotal + GST

4. Invoice Generation Module

This module generates the final bill for the customer.

The bill includes:

• Item name
• Quantity
• Price
• Subtotal
• GST amount
• Final bill amount

Example Bill Output:

------ HOTEL BILL ------

Item	Qty	Price
Chicken Biryani	2	400
Cold Drink	2	60

Subtotal	460
GST (5%)	23
Total	483

Thank You Visit Again!

5. Exit Module

This module allows the user to exit the application safely after completing the billing process.

When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Hotel & Restaurant Billing System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of restaurant operations.

The main problems identified in manual billing systems are:

- Manual calculations may lead to billing errors.
- Preparing bills manually takes more time.
- Difficult to handle multiple customer orders efficiently.
- Lack of automated records for transactions.

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies restaurant billing operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Menu Display Module
• Order Selection Module
• Bill Calculation Module
• Invoice Generation Module
• Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of menu-driven logic
• Writing code for bill calculation
• Handling user input
• Displaying the final bill

Core Java concepts such as **classes, objects, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Select menu item	Item displayed correctly
Enter quantity	Correct quantity accepted
Calculate bill	Correct total displayed
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected food items and quantities.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Hotel & Restaurant Billing System**:

1. Start the program.
2. Display the list of food items with prices.
3. Ask the user to select a food item.
4. Enter the quantity of the selected item.
5. Calculate the total cost of the item.
6. Add the cost to the total bill.
7. Repeat steps 3–6 for additional items if required.

8. Calculate the GST or tax amount.
9. Calculate the final bill amount.
10. Display the final bill to the user.
11. End the program.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Display Menu
↓
Select Item
↓
Enter Quantity
↓
Calculate Bill
↓
Display Invoice
↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import javax.swing.*;
import java.util.*;

class MenuItem {
  int id;
  String name;
  double price;

  MenuItem(int id, String name, double price) {
    this.id = id;
    this.name = name;
    this.price = price;
  }
}

class OrderItem {
  MenuItem item;
  int quantity;

  OrderItem(MenuItem item, int quantity) {
    this.item = item;
    this.quantity = quantity;
  }

  double getTotal() {
    return item.price * quantity;
  }
}

public class RestaurantPopupSystem {

  static List<MenuItem> menu = new ArrayList<>();
```

```
  static List<OrderItem> orders = new ArrayList<>();

  public static void main(String[] args) {

    // Menu initialization
    menu.add(new MenuItem(1, "Burger", 120));
    menu.add(new MenuItem(2, "Pizza", 250));
    menu.add(new MenuItem(3, "Pasta", 180));
    menu.add(new MenuItem(4, "Coke", 50));

    int option;

    do {
      String menuText = "------ MENU ------\n";
      for (MenuItem item : menu) {
        menuText += item.id + ". " + item.name + " - ₹" + item.price +
"\n";
      }

      String input = JOptionPane.showInputDialog(menuText + "\nEnter
Item ID:");
      if (input == null) break;

      int id = Integer.parseInt(input);
      MenuItem selected = findItem(id);

      if (selected != null) {
        String qtyInput = JOptionPane.showInputDialog("Enter
quantity:");
        int qty = Integer.parseInt(qtyInput);

        orders.add(new OrderItem(selected, qty));
        JOptionPane.showMessageDialog(null, "Item added!");
      } else {
        JOptionPane.showMessageDialog(null, "Invalid item!");
      }

      option = JOptionPane.showConfirmDialog(null,
          "Add more items?", "Continue",
          JOptionPane.YES_NO_OPTION);
```

```
    } while (option == JOptionPane.YES_OPTION);

    showBill();
  }

  static MenuItem findItem(int id) {
    for (MenuItem item : menu) {
      if (item.id == id) return item;
    }
    return null;
  }

  static void showBill() {
    double total = 0;
    String bill = "------ BILL ------\n";

    for (OrderItem order : orders) {
      double itemTotal = order.getTotal();
      total += itemTotal;

      bill += order.item.name + " x " + order.quantity +
          " = ₹" + itemTotal + "\n";
    }

    double tax = total * 0.05;
    double finalAmount = total + tax;

    bill += "\nSubtotal: ₹" + total;
    bill += "\nTax (5%): ₹" + tax;
    bill += "\nTotal: ₹" + finalAmount;
    bill += "\n\nThank You! 😊";

    JOptionPane.showMessageDialog(null, bill);
  }
}
```

```
import javax.swing.*;
import java.util.*;

class MenuItem {
  int id;
  String name;
```

```
    double price;

    MenuItem(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }
}

class OrderItem {
    MenuItem item;
    int quantity;

    OrderItem(MenuItem item, int quantity) {
        this.item = item;
        this.quantity = quantity;
    }

    double getTotal() {
        return item.price * quantity;
    }
}

public class RestaurantPopupSystem {

    static List<MenuItem> menu = new ArrayList<>();
    static List<OrderItem> orders = new ArrayList<>();

    public static void main(String[] args) {

        // Menu initialization
        menu.add(new MenuItem(1, "Burger", 120));
        menu.add(new MenuItem(2, "Pizza", 250));
        menu.add(new MenuItem(3, "Pasta", 180));
        menu.add(new MenuItem(4, "Coke", 50));

        int option;

        do {
            String menuText = "------ MENU ------\n";
            for (MenuItem item : menu) {
```

```
            menuText += item.id + ". " + item.name + " - ₹" + item.price +
"\n";
        }

        String input = JOptionPane.showInputDialog(menuText + "\nEnter
Item ID:");
        if (input == null) break;

        int id = Integer.parseInt(input);
        MenuItem selected = findItem(id);

        if (selected != null) {
            String qtyInput = JOptionPane.showInputDialog("Enter
quantity:");
            int qty = Integer.parseInt(qtyInput);

            orders.add(new OrderItem(selected, qty));
            JOptionPane.showMessageDialog(null, "Item added!");
        } else {
            JOptionPane.showMessageDialog(null, "Invalid item!");
        }

        option = JOptionPane.showConfirmDialog(null,
                "Add more items?", "Continue",
                JOptionPane.YES_NO_OPTION);

    } while (option == JOptionPane.YES_OPTION);

    showBill();
  }

  static MenuItem findItem(int id) {
    for (MenuItem item : menu) {
        if (item.id == id) return item;
    }
    return null;
  }

  static void showBill() {
    double total = 0;
    String bill = "------ BILL ------\n";
```

```
    for (OrderItem order : orders) {
        double itemTotal = order.getTotal();
        total += itemTotal;

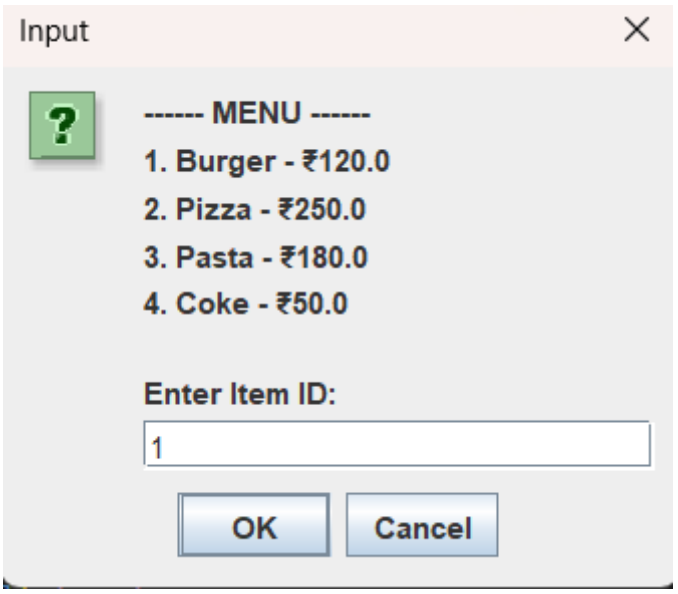
        bill += order.item.name + " x " + order.quantity +
                " = ₹" + itemTotal + "\n";
    }

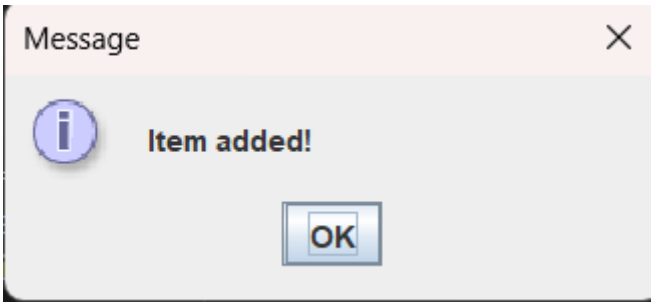
    double tax = total * 0.05;
    double finalAmount = total + tax;

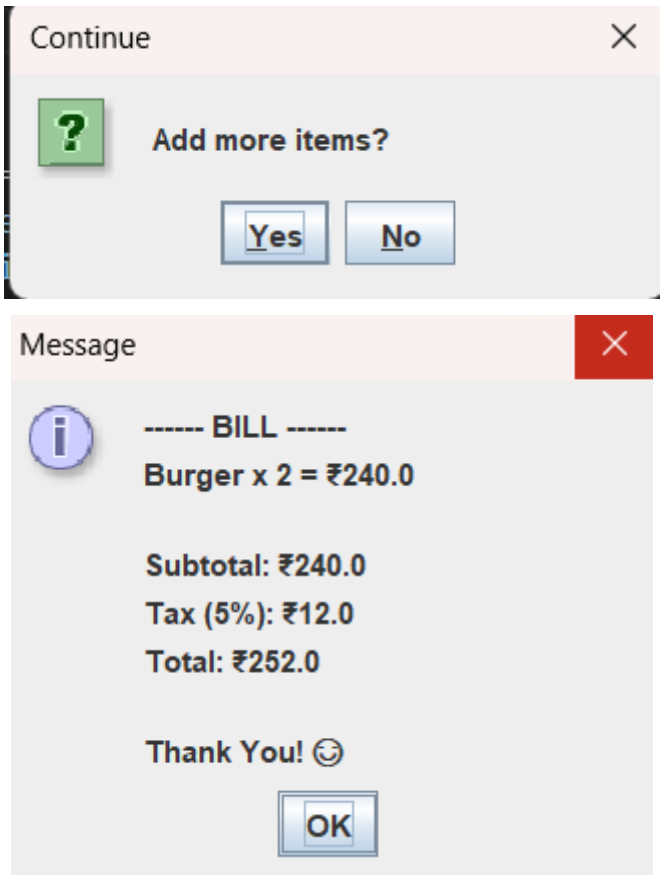
    bill += "\nSubtotal: ₹" + total;
    bill += "\nTax (5%): ₹" + tax;
    bill += "\nTotal: ₹" + finalAmount;
    bill += "\n\nThank You! 😊";

    JOptionPane.showMessageDialog(null, bill);
  }
}
```

6. INPUT AND OUTPUT SCREENS






Continue
Add more items?
Yes
No
Message
------ BILL ------
Burger x 2 = ₹240.0
Subtotal: ₹240.0
Tax (5%): ₹12.0
Total: ₹252.0
Thank You!
OK

7. DISCUSSION & CONCLUSION

7.1 Discussion

The**Restaurant Order Processing and Billing** System was developed to automate the billing process in restaurants and hotels. The system allows users to select food items, enter quantities, and generate bills automatically.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs bill calculations and generates accurate invoices.

The system was tested with different inputs to ensure that it performs the required calculations correctly. The results show that the system reduces manual effort and improves billing efficiency.

7.2 Conclusion

The **Restaurant Order Processing and Billing System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world business application.

The system automates the billing process, reduces calculation errors, and provides faster service to customers. It simplifies restaurant billing operations by generating accurate invoices based on customer orders.

This project helps students understand how Java programming can be used to develop practical software systems. The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, and online payment systems.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com